

JUST AUTOS

JA Portal — Handover

How the portal is built, where it runs, every connection it has,
and the operating procedures for every module.

Compiled 20 August 2026 · justautos.app · Internal — not for distribution outside Just Autos

JA Portal — Full Handover Document

How the portal is built, where it runs, every connection it has (and how to set up or change them), and operating procedures for every module.

Compiled 20 August 2026 from the live codebase. Supersedes `03_SYSTEM_OVERVIEW.md` / `05_INTEGRATIONS.md` (May 2026) where they conflict.

1. What the portal is

An internal management platform for **Just Autos**, covering two MYOB business entities:

- **JAWS** — Just Autos Wholesale (distribution arm, holds stock, ~14 distributors across Australia)
- **VPS** — Vehicle Performance Solutions (the workshop entity; runs on **Mechanics Desk**, which remains the system of record — see §7.2)

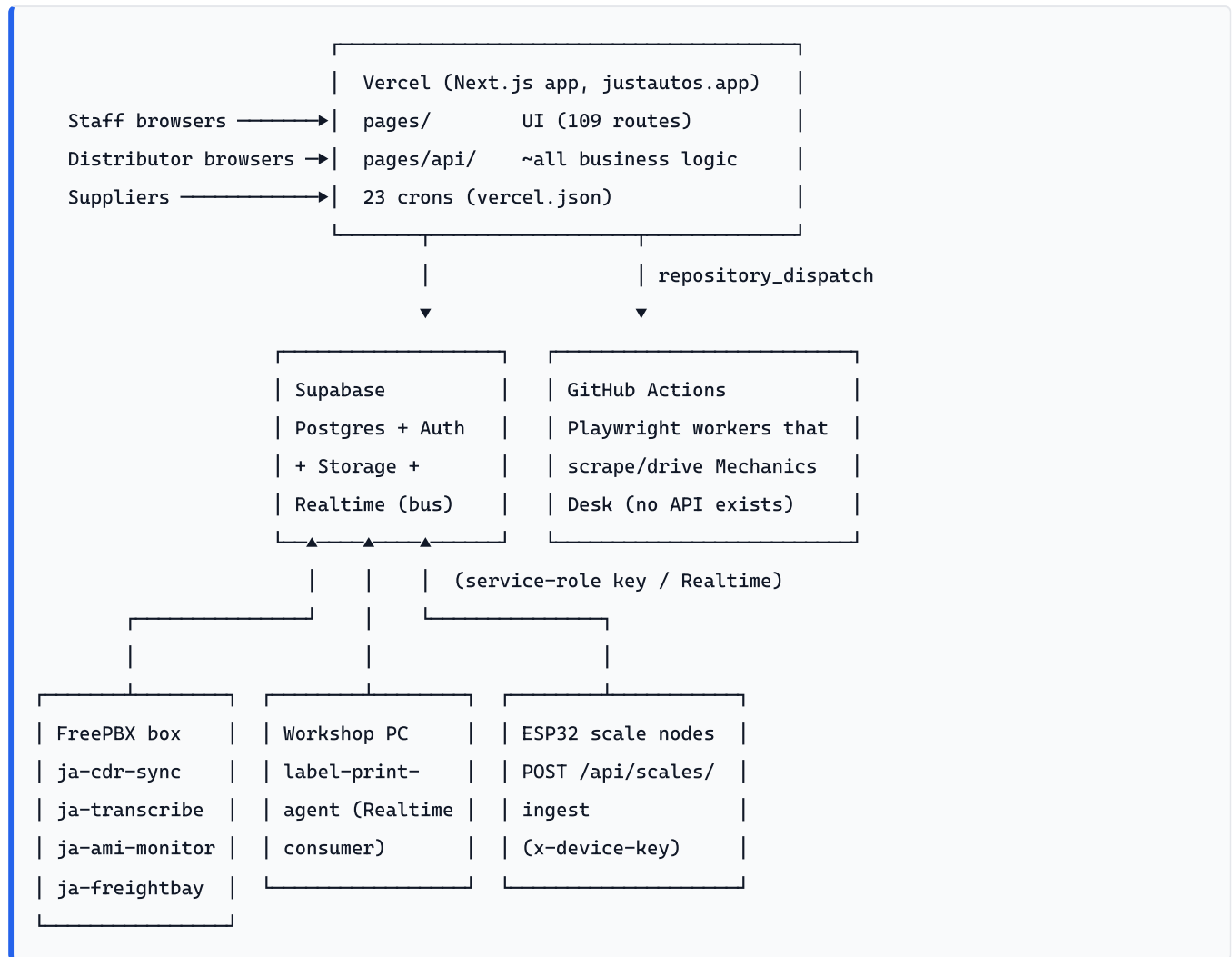
It is one Next.js application that contains: a **staff portal** (dashboards, workshop management, CRM, AP automation, calls coaching, reporting), a **distributor-facing B2B portal** (catalogue, checkout, orders, tune jobs, training) that went live in July 2026, a **supplier portal** (read-only stock wall), and a large fleet of **background automation** (23 Vercel crons, 16 GitHub Actions workflows, and several on-premise agents).

Production URL	<code>https://justautos.app</code>
Repo	<code>ChrisJustAutos/JA-Portal</code> , branch <code>main</code>
Hosting	Vercel (auto-deploys every push to <code>main</code>)
Database	Supabase project <code>qtiscbvhlvdvafwtdtcd</code> (<code>https://qtiscbvhlvdvafwtdtcd.supabase.co</code>)
Framework	Next.js 14.2.5, pages router , TypeScript, React 18
Key packages	<code>@supabase/supabase-js</code> , <code>@anthropic-ai/sdk</code> (mostly raw fetch is used), <code>exceljs</code> / <code>xlsx</code> , <code>@react-pdf/renderer</code> , <code>pdf-lib</code> , <code>leaflet</code> , <code>reactflow</code> , <code>sip.js</code> , <code>web-push</code> , <code>playwright</code> (dev, for GH Actions workers)

People

Person	Role
Chris Russell	Operations Manager — portal owner/direction
Nat	Accountant — chart of accounts, MYOB reconciliation sign-off
Matt Ashley	Technical — MYOB API, integrations
Matt H	Operations / Sales
Amanda	Accounts Payable — works in <code>/ap</code> daily
Laura	Director
Ryan	Receives the Monday 7am Weekly Sales Recap email

2. Architecture at a glance



External SaaS the Vercel app talks to directly: **MYOB AccountRight** (both entities, direct OAuth), **Xero** (migration target, foundation live), **Stripe** (B2B standalone account + JAWS read-only accounts), **Microsoft Graph** (mailbox webhooks + mail), **Resend** (email), **ClickSend** (SMS), **Slack** (bot + webhooks), **Monday.com**, **ActiveCampaign** (being replaced by CRM module), **MachShip** (freight), **Anthropic API** (18 LLM call sites), **Deepgram** (indirectly — runs on the PBX host), **Google Places / Nominatim** (geocoding).

Design conventions (must-follow)

- **API routes are `.ts`, never `.tsx`** — a `.tsx` in `pages/api/` crashes module load (generic 500 HTML).
- **Shared UI kit is mandatory:** `lib/ui/theme` tokens (`T` — CSS variables; use `alpha()`, never alpha-suffixed tokens) + `components/ui` + Feedback hooks. No `alert()` / `confirm()` browser dialogs. The B2B portal has its own separate "Alloy" kit at `components/b2b/ui.tsx` (one accent, $\geq 12\text{px}$ type, 44px touch targets).
- **Server Supabase clients** are constructed inline per route with the service-role key (module-level memo `_sb`). There is deliberately no shared server client module — follow the local pattern.
- **Long work (>~30s) never runs on Vercel** — anything browser-based or slow goes to GitHub Actions (dispatch pattern) or is chunked.
- **PostgREST embeds on workshop tables must use the `!customer_id` hint** (e.g. `workshop_customers!customer_id(...)`) or the query 500s on ambiguous relationships.

- `b2b_distributor_users` is not unique per email/auth_user_id (multi-site memberships) — never `maybeSingle()` on it.
- An **update notifier** compares the client bundle against `/api/version` and shows a "new version — Reload" banner after each deploy.

3. Environments, deployment & dev workflow

Deploy

1. Commit to `main` and **push** (convention: always push immediately after committing — never leave local-only commits).
2. Vercel builds and deploys production automatically. There is **no staging branch**; PR preview URLs exist if wanted.
3. Function logs: Vercel dashboard → Deployments → deploy → Functions tab.

Error fingerprints: generic 500 HTML page = module-load crash (bad import / `.tsx` in `api/` / syntax error); JSON error = handler crash; empty response/timeout = function `maxDuration` exceeded (per-route overrides live in `vercel.json` → `functions`).

Database migrations (SOP)

- Migrations live in `migrations/NNN_description.sql` (196 files, `002` – `195`; note `148` and `153` are each duplicated — sequence is a convention, not a key. Next number: **196**).
- There is **no migration runner**. The procedure is: write the SQL file in `migrations/`, apply it to the live DB via the **Supabase MCP** `apply_migration` tool (project `qtiscbvhlvdvafwtdtdc`) **before** pushing code that depends on it, then commit both.
- The repo file is the source-of-truth record; the MCP apply is what actually changes the DB.

Local dev

`npm run dev` with a `.env.local` carrying at minimum the Supabase URL/keys. Most features hit live external services, so local dev is mainly for UI work; be careful with anything that writes to MYOB/Stripe.

Remote/mobile ops

A workshop MSI laptop runs OpenSSH + Tailscale + Claude Code; Chris's phone (Tailscale + Termius) can SSH in from anywhere. This is also the **only interactive route to the FreePBX box** (Tailscale IP `100.82.97.46`).

4. Auth & access model

Four separate authentication schemes coexist. They never overlap.

4.1 Staff portal

- Supabase Auth session, cookie `ja-portal-access-token`; identity in `user_profiles`. Opt-in TOTP 2FA (server-side enforced).
- **Six roles** (`lib/permissions.ts` is the source of truth): `admin`, `manager`, `sales`, `workshop`, `accountant`, `viewer`. (`lib/auth.ts` has a stale 5-role type missing `workshop` — known drift.)

- Roles map to `view:* / edit:* / admin:*` permission strings via `ROLE_PERMISSIONS`. Pages gate with `requirePageAuth(ctx, 'permission');` API routes with `withAuth(permission, handler) / requireAdmin`.
- Two per-user allowlists narrow further: `user_profiles.visible_tabs` (which nav tabs/apps a user sees) and `visible_report_tabs` (which Reports sub-tabs).
- **User management:** `/settings?tab=users` — invite (Supabase invite email → `/reset-password?welcome=1`), change role, deactivate, delete. Audit log at `/settings?tab=audit`.
- Session persistence: `SessionKeeper` silently re-syncs cookies from `localStorage` (never on auth pages) and login pages silently resume valid sessions; deep links use `?next=`.

4.2 Distributor (B2B) portal

- Separate cookies `ja-b2b-access-token / ja-b2b-refresh-token`; identity in `b2b_distributor_users`. Gate: `requireB2BPageAuth`.
- Login = email+password with optional TOTP; also magic-link and signed invite tokens (`/b2b/welcome?t=...` — scanner-proof: opening does nothing, only the human submitting the set-password form activates).
- Users are `owner` or `member` per distributor; one auth user can belong to **multiple distributors** (account switcher, `ja-b2b-dist` cookie).
- Admin **preview mode:** signed `b2b_preview` token from the admin side renders the portal as a distributor with all non-GET requests blocked.
- `checkoutEnabled` per distributor = browse-only kill switch.

4.3 Supplier portal

Separate again: `b2b_supplier_users`, `requireSupplierPageAuth`, single read-only page (`/b2b/supplier`).

4.4 Machine auth

- **Service tokens** (`lib/service-auth.ts`): SHA-256 hashed rows in `service_tokens`, presented as `X-Service-Token` header (deliberately not `Authorization` to avoid clashing with Supabase JWTs). Scoped (`stocktake:write`, `calls:monitor`, `ap:admin`, `reports:read`, `upload:job-report`). Used by GitHub Actions workers and the PBX AML agent. Managed via `/api/admin/service-tokens`.
- **Signed capability tokens** in URLs for login-less pages: `/tune-jobs?token=` (distributor-scoped weekly reminder), `/order-action` (admin Book Freight email button).
- **Device keys:** ESP32 scale modules authenticate to `/api/scales/ingest` with `x-device-key` matched to `scale_devices`.
- **MCP tokens:** `jap_...` personal tokens (hashed in `mcp_tokens`) or OAuth 2.1 for the read-only Claude connector at `/api/mcp`.
- **Cron auth:** `Authorization: Bearer $CRON_SECRET` (some handlers also accept the `vercel-cron` user-agent, and some accept a logged-in staffer with the right permission for manual runs).

5. Connections & integrations

5.1 Where credentials live (read this first)

`lib/integration-config.ts` implements a **DB-first resolver**: `getIntegration(key)` = `integration_settings` DB row → env var of same name → `''`. Values cache in-process for 30s. **Always use this resolver, not `process.env`, for the managed keys.**

Integration	Stored where	Change without redeploy?
ClickSend, Resend, CRM intake token, Xero app creds , accounting provider switch	<code>integration_settings</code> DB (env fallback)	Yes — Settings → Connections → Integrations; live in ~30s
MYOB app creds, Graph, Monday, ActiveCampaign, Stripe, Slack, Anthropic, VAPID, Supabase, GitHub dispatch	Vercel env vars only	No — edit env + redeploy
MYOB / Xero OAuth tokens	<code>myob_connections</code> / <code>xero_connections</code> tables	Yes — re-run connect flow
MachShip token	<code>b2b_freight_carrier_connections.credentials</code> JSONB	Yes — Admin B2B → Settings
Service tokens / MCP tokens / scale device keys	<code>service_tokens</code> / <code>mcp_tokens</code> / <code>scale_devices</code>	Yes — respective admin UIs

Admin surfaces:

- **Settings** → **Connections** (`/settings?tab=connections`) — three sub-tabs: **Integrations** (edit DB-managed credentials, test SMS/email, rotate CRM intake token, Xero connect), **Health** (live status board), **MYOB Connection** (connect/select company files).
- `/admin/connections` — standalone auto-refreshing health page (⚠ no SSR auth gate — relies on API-level gating).

5.2 MYOB AccountRight (primary accounting — both entities)

- **Code**: `lib/myob.ts` (OAuth + `myobFetch`), `lib/myob-reporting.ts` (replaced CData 2026-07-14 — reporting **must fetch all 4 invoice types**), plus per-module writers (`ap-myob-bill`, `b2b-myob-invoice`, `workshop-myob-invoice`, `stripe-myob-sync`, ...).
- **App creds (env only)**: `MYOB_CLIENT_ID`, `MYOB_CLIENT_SECRET`, `MYOB_REDIRECT_URI`, `MYOB_SCOPE`.
- **Tokens**: `myob_connections` table, one row per label `JAWS` / `VPS`. Access ~20 min (auto-refresh), refresh ~1 year but **must be exercised regularly** — a multi-week quiet spell can kill the connection.
- **Connect / re-auth SOP**: Settings → Connections → MYOB Connection → Connect (or `GET /api/myob/auth/connect?label=JAWS|VPS`) → consent → pick company file → if the file is in legacy auth mode, set company-file username/password. Test with `/api/myob/test/invoice?label=VPS`.
- **Gotchas** (hard-won, in code comments):
 - Two company-file auth modes — SSO (bearer only) vs legacy (`x-myobapi-cftoken`); header only sent when `company_file_username` is set.
 - **Never page with bare `$top` / `$skip`** — no stable ordering, rows get skipped. Follow `NextPageLink`, page size 400.

- Every call is logged to `myob_api_log`; the insert is awaited in `finally` (fire-and-forget logs were killed on timeout).
- **IsTaxInclusive matters for cent-exact invoices** — B2B invoices are pushed inc-GST with checkout-exact line pricing.
- No P&L endpoint exists in AccountRight (the old CData P&L panels were retired).

5.3 Xero (migration target — foundation live 2026-08-05)

- **Code:** `lib/xero.ts`, `lib/accounting/xero-adapter.ts`; provider switch in `lib/accounting-provider.ts`.
- **App creds (DB-managed):** `XERO_CLIENT_ID`, `XERO_CLIENT_SECRET`, `XERO_REDIRECT_URI` — editable in the Integrations tab.
- **Setup SOP:** create a Web App at `developer.xero.com` with the redirect URI shown on the card → paste ID/secret → Save → "Connect Xero" (`/api/xero/auth/start`) → one consent covers both orgs → map tenants to `VPS / JAWS` via `/api/xero/connections` PATCH → ping to verify.
- **Provider switch:** `accountingProvider(entity, module?)` resolves `ACCOUNTING_PROVIDER_{ENTITY}_{MODULE}` → `ACCOUNTING_PROVIDER_{ENTITY}` → default `'myob'`. Entity-wide keys are in the Integrations UI; per-module overrides (AP, STATEMENTS, LETTERS, WORKSHOP, INVENTORY, B2B, REPORTING, STRIPE, BANK) are env-only. After cutover MYOB stays connected read-only for history.
- **Critical gotcha: Xero refresh tokens are single-use and rotate on every refresh** — the new token must be persisted or the connection dies. Refresh is serialised in-process; a cross-instance race shows as `invalid_grant` and self-heals next run. Granular scopes only (bills live under `accounting.invoices`); every call needs the `Xero-tenant-id` header.

5.4 Stripe

- **B2B checkout (standalone account `acct_1TrRkZ...`, LIVE keys):** `lib/stripe.ts` (raw fetch, no SDK). Env: `STRIPE_SECRET_KEY`, `STRIPE_WEBHOOK_SECRET`. Webhook POST `/api/b2b/stripe/webhook` (event `checkout.session.completed`; `bodyParser: false` is required for signature verification — do not remove). Idempotent; if the MYOB writeback fails it still 200s and parks the error on `b2b_orders.myob_write_error` for manual retry from the admin order page. Payment methods live: **card (with surcharge), PayTo, BECS direct debit** (BECS settles later — a settle gate + `/api/cron/b2b-payment-check` confirm before fulfilment milestones).
- **JAWS reconciliation accounts (read-only):** `lib/stripe-multi.ts`, env `STRIPE_SECRET_KEY_JAWS_JMACX`, `STRIPE_SECRET_KEY_JAWS_ET`, labels registered in `STRIPE_ACCOUNT_LABELS`. Used by `/stripe-myob` push tool and payout reconciliation.

5.5 Microsoft Graph (mail)

- **Auth:** app-only client credentials. Env: `GRAPH_TENANT_ID`, `GRAPH_CLIENT_ID`, `GRAPH_CLIENT_SECRET` (+ webhook URLs and ~15 mailbox-assignment vars — see `lib/microsoft-graph.ts` and §22 of the integrations inventory). Azure app needs **Mail.Read, Mail.ReadWrite, Mail.Send with admin consent** (Mail.Send 403 until granted — this bit us).
- **Webhooks:** `/api/webhooks/graph-mail` (Pipeline A — quote PDFs from rep mailboxes), `/api/webhooks/graph-jobreport-mail` (nightly MD WIP report).

- **Subscription lifecycle SOP:** subscriptions max out at ~70.5h. The cron `/api/cron/renew-graph-subscriptions` (every 6h) extends anything within 24h of expiry. **Recreating a dead/failed subscription is NOT automated** — re-run `POST /api/admin/setup-graph-subscriptions?key=$GRAPH_ADMIN_SETUP_SECRET` (idempotent).
- **Adding a rep mailbox** touches three places: `lib/agents.ts` (owner map), `MAILBOX_FOR_NAME` in `pages/api/cron/health-check.ts`, and re-running the setup endpoint.
- Inbox-driven pipelines that *poll* Graph rather than subscribe: AP inbox pull, AP auto-entry, AP statement watch, tune-jobs receipt scan, drop-ship confirmations, letter watch.

5.6 Email (Resend) & SMS (ClickSend)

- **Resend** (`lib/email.ts`): DB-managed keys `RESEND_API_KEY`, `RESEND_FROM`, `RESEND_CAMPAIGN_FROM`, `RESEND_REPLY_TO`, `RESEND_WEBHOOK_SECRET`. If unset, **silently falls back to Graph sendMail**. Delivery/open/bounce webhook: `/api/crm/resend-webhook?key=...`. Per-staff Reply-To comes from `user_profiles.reply_to_email`.
- **ClickSend** (`lib/clicksend.ts`): DB-managed `CLICKSEND_USERNAME`, `CLICKSEND_API_KEY`, `CLICKSEND_FROM`. Returns `clicksend_not_configured` harmlessly until set. Test buttons for both live in the Integrations tab.

5.7 Slack

- **Incoming webhooks** (one env URL per channel): `SLACK_WEBHOOK_URL` plus `SLACK_WEBHOOK_{JAWS_PAYMENTS,VPS_PAYMENTS,JAWS_PAYOUTS,AP_VPS}` and per-tenant B2B order webhook in `b2b_settings`.
- **Bot** (`lib/slack-bot/*`): env `SLACK_BOT_TOKEN`, `SLACK_SIGNING_SECRET`, `SLACK_ALLOWED_CHANNEL_IDS`, `SLACK_PARTS_ONLY_CHANNEL_IDS` (parts-only channels get exactly one stock-availability answer per top-level message, no thread replies, no clarifying questions), `SLACK_PARTS_CONTACT`. Single endpoint `POST /api/slack/ask` handles events, mentions, `/ask`, and interactive buttons; HMAC verified; acks in <3s then continues via `waitUntil`.
- The parts bot answers from `md_stock_cache`, refreshed every 30 min by the `md-stock-sync` GitHub Action.

What we post where. Slack is the business's alerting surface — most automation reports into it rather than email, so a dead webhook is a silent failure. The channels built so far:

Channel	What lands there	Source
#jaws-orders	B2B distributor orders, drop-ship PO confirmations and ETAs	<code>lib/b2b-order-pipeline.ts</code> , drop-ship confirm cron
#jaws-payments / #vps-payments	Payments received, per entity	<code>SLACK_WEBHOOK_{JAWS,VPS}_PAYMENTS</code>
#jaws-invoices / #vps-invoices	Invoice events per entity	webhook

Channel	What lands there	Source
#sales-coaching	Per-call coaching cards, plus the weekly team coaching summary (Mon 07:00)	PBX slack-poster.js
#customer-feedback-negative	Calls flagged as a poor customer experience, with the fault theme. Matt's standing ask is the top-10 recurring faults off this channel. The automation that posts here is currently switched OFF (CALL_CONCERNS_ENABLED).	calls analysis
#customer-feedback-positive	The same in reverse — calls that went well	calls analysis
#parts-room	Parts Office camera motion, nights and weekends only	ja-partsroom on the PBX box
Freight bay	Intrusion bursts from the freight-bay NVR (snapshot sequence)	ja-freightbay on the PBX box — Slack-only since 2026-07-15
AP channel (SLACK_WEBHOOK_AP_VPS)	Supplier-invoice exceptions and suspected double-ups (🔄)	AP auto-entry
#ja-portal-queries	Ask-the-portal bot conversations	lib/slack-bot/*

⚠ Webhook URLs are per-channel env vars. If a channel goes quiet, check the env var before assuming the feature broke — a rotated or deleted webhook fails silently.

5.8 Monday.com & ActiveCampaign (legacy CRM pair — being replaced by the CRM module)

- **Monday** (lib/monday-followup.ts is the shared GraphQL client): env MONDAY_API_TOKEN , MONDAY_BUTTON_SECRET . Five rep quote boards (Dom 5025942308, Kaleb 5025942316, Graham 5026840169, James 5025942292, Tyronne 5025942288). Pipeline A is match-only (never creates items). Button callback: /api/monday/fetch-call-notes?key=... .
- **Quote-channel automations (rebuilt 2026-08-19/20)**: a 3/7/14-day follow-up cadence driven by an FU Stage column (1→2→3; "Follow Up Done" advances the stage, resets Contact Attempts, hides the item in Quote - Pending and re-surfaces it when the date arrives). RLMNA counts attempts: **5 tries in Quote - Lead RLMNA → Quote Not Issued, 3 tries in Quote - Follow up RLMNA → Quote Lost**. After the third follow-up the owner is notified that the decision must be Won or Lost.
 - **Two landmines, both fixed 2026-08-20, worth knowing if the boards misbehave again.** (1) Graham's legacy RLMNA - Step 1 had no group condition and his Follow up RLMNA - Step 1 incremented the counter twice, so one RLMNA click added +3 attempts and lost the quote on the first click; rebuilt as new-builder workflows 1940255686 / 1940255690 with the move ordered *before* the increment. (2) The Follow-Up-Done handlers are gated on FU Stage is exactly 1/2/3 , so a **blank** FU Stage makes the click do nothing at all, silently — 132 items (mostly the On Hold cohort the migration skipped) were backfilled to stage 1.

- **MCP cannot delete or deactivate Monday automations** (`USER_UNAUTHORIZED` , even for the account owner — the connector lacks the scope). The workaround is always: create a corrected version, add the bad one to a manual delete list. `create_automation` is also **nondeterministic** — always re-dump with `list_automations` and check the block config; phrase status triggers as "changes to X (from any previous status)" or it will set the from-status to the same label and the workflow can never fire.
- **ActiveCampaign** (`lib/activecampaign.ts`): env `ACTIVECAMPAIGN_API_URL` , `ACTIVECAMPAIGN_API_KEY` , owner map etc. **Landmine documented in code: `filters[phone]` is a partial match and matches everything on an empty query** — every match must be re-verified by exact digit comparison (this mis-attributed a contact to 26 customers in production once).

5.9 MachShip (B2B freight)

- **Code:** `lib/b2b-machship.ts` + `lib/b2b-freight-carriers.ts` (provider registry — Shippit/Starshipit/AusPost/Sendle slots exist but only MachShip is implemented).
- **Credential:** NOT env — `b2b_freight_carrier_connections` row `provider='machship'` , `JSONB api_token` , header `token:` (not Bearer). Change it in Admin B2B → Settings; effective next call.
- Base `https://live.machship.com` (no sandbox host — test vs live is a property of the token's MachShip user). **Manifesting needs `companyId` , obtained via a consignment GET.** Booking runs with `maxDuration: 120` ; consignments that die at MachShip park as `consignment_missing` ; `/api/cron/b2b-freight-poll` refreshes status/ETA every 30 min.

5.10 Phones: FreePBX + Deepgram + live monitoring

The portal never calls Asterisk or Deepgram directly — an on-PBX host (CentOS 7, Tailscale `100.82.97.46`) runs four workers and **Supabase is the bus**:

PBX worker	What it does
<code>ja-cdr-sync</code> (<code>sync.js</code>)	CDR → Supabase <code>calls</code> every 5 min (service-role key). Includes park-pickup CDR fix (2026-08-07).
<code>ja-transcribe</code> (<code>transcribe.js</code>)	New calls → Deepgram (<code>nova-2-phonecall</code> , <code>en-AU</code>) → <code>call_transcripts</code> . Deepgram key lives on the PBX host.
<code>ja-ami-monitor</code>	Live channel snapshots → <code>POST /api/calls/live/agent/snapshot</code> (~2s, <code>X-Service-Token</code> scope <code>calls:monitor</code>); drains <code>call_monitor_events</code> for Listen/Whisper/Barge and click-to-dial originate.
<code>ja-freightbay</code> / <code>ja-partsroom</code>	Hikvision NVR intrusion events → Slack snapshot bursts + Yealink ring. Node 16 only (glibc). Doesn't touch the portal API.

Call coaching and notes — what this actually gives the business. This is the biggest piece of bespoke work on the phone system, so it is worth stating plainly what it produces:

1. **Every call is recorded and transcribed** — `ja-cdr-sync` lands the CDR, `ja-transcribe` sends the audio to Deepgram (`nova-2-phonecall` , `en-AU`) and stores the transcript. Nothing is manual.
2. **Every call is scored against a rubric for its type** — a sales enquiry, a service booking, a parts call and a pass-off are judged on different things, so the rubrics are per call type (`call_type_rubrics` , editable at Settings → Call Coaching).

3. **Coaching is attributed to the advisor who actually handled it**, identified from the transcript rather than the extension — that is what makes the leaderboard trustworthy when calls get transferred or picked up from park.
4. **Coaching cards post to #sales-coaching** per call, and a **team summary posts Monday 07:00**.
5. **Call notes flow back to the quote boards** — the "Fetch Call Notes" button on a Monday item calls `/api/monday/fetch-call-notes`, which pulls that customer's call history and notes onto the item, so a rep picking up a follow-up can see what was last said without hunting for the recording.
6. **Sentiment / objections / conversion** are surfaced on `/calls` as tabs, and calls scoring below 40 are flagged for attention.

Analysis runs on the PBX (`/opt/ja-cdr-sync/analyse.js` + `slack-poster.js`); the portal's `/api/cron/calls-analyse` equivalent is **gated off by default** (`CALLS_ANALYSIS_ENABLED`) and must stay off while the PBX loop runs — both running means double-scoring and double-posting. Browser softphone (SIP.js over WSS) config is env `NEXT_PUBLIC_FREEPBX_WSS_URL` etc.; TURN needed on mobile networks.

5.11 Anthropic / Claude API

Env `ANTHROPIC_API_KEY`; raw fetch to `/v1/messages`. 18 call sites: AP extraction & statement parsing, quote PDF extraction, call analysis/coaching/concerns/weekly report, follow-up summaries, workshop-map weekly narrative, B2B drop-ship confirmation reading, training-course generation, tune-job receipt extraction, report narratives, sales-recap flags, Slack bot, portal chat. Most models overridable per-feature via env (`*_MODEL` vars).

5.12 Everything else

- **Web Push**: `NEXT_PUBLIC_VAPID_PUBLIC_KEY` / `VAPID_PRIVATE_KEY` (watch for whitespace in env values — known gotcha); dormant no-op when unset.
- **MCP connector**: portal is a read-only MCP server at `/api/mcp` (personal `jap_` tokens from Settings, or OAuth 2.1 with dynamic client registration).
- **Google Places** (`NEXT_PUBLIC_GOOGLE_PLACES_API_KEY`, client-exposed — must be referrer-restricted) for workshop address autocomplete; **Nominatim** (keyless, throttled ~1 rps) geocodes tune-job addresses; an offline AU postcode dataset powers the workshop map.
- **GitHub dispatch**: `GH_DISPATCH_TOKEN` (`actions:write`) lets portal routes fire `repository_dispatch` events to run MD workers on demand.
- **Website lead intake**: `POST /api/crm/intake` guarded by `x-crm-token` = DB-managed `CRM_INTAKE_TOKEN` (rotate from the Integrations tab).
- **CData MCP**: decommissioned 2026-07-14. `lib/cdata.ts` is dead code; health check hardcoded green/deprecated.
- **Base URL sprawl**: seven overlapping env vars (`NEXT_PUBLIC_APP_URL`, `NEXT_PUBLIC_SITE_URL`, `NEXT_PUBLIC_BASE_URL`, `NEXT_PUBLIC_PORTAL_URL`, `PORTAL_PUBLIC_URL`, `APP_BASE_URL`, `JA_PORTAL_BASE_URL`, `B2B_PUBLIC_BASE_URL`) — keep them consistent; mismatches silently break OAuth redirects and email links.

6. Scheduled automation

6.1 Vercel crons (23 — `vercel.json` ; schedules are UTC; Brisbane = UTC+10)

Cron	Brisbane time	Purpose
<code>/api/distributors/refresh-cache</code>	02:00 daily	Warm <code>distributors_cache</code> (FY figures + last 13 months)
<code>/api/cron/sync-followups</code>	every 5 min	Call follow-ups → Claude summary → AC contact + Monday push (<code>FOLLOWUP_SYNC_ENABLED</code> kill switch)
<code>/api/cron/calls-analyse</code>	every 5 min	Portal-side call coaching sweep — off by default (<code>CALLS_ANALYSIS_ENABLED</code>); PBX box currently does this
<code>/api/cron/calls-weekly-report</code>	Mon 07:00	Weekly coaching narrative → <code>#sales-coaching</code>
<code>/api/cron/workshop-map-weekly</code>	Mon 07:10	Quotes/jobs geography report → email Matt (cc Ryan, Chris)
<code>/api/cron/renew-graph-subscriptions</code>	every 6 h	Extend Graph mailbox subscriptions expiring <24h
<code>/api/cron/health-check</code>	every 5 min	The 21 integration health checks → <code>integration_health</code>
<code>/api/cron/b2b-freight-poll</code>	every 30 min	MachShip status/ETA poll on open consignments
<code>/api/cron/workshop-reminders</code>	every 15 min	Queue + send service/regio reminder SMS (respects <code>sms_enabled</code>)
<code>/api/cron/notifications-sweep</code>	every 15 min	Bell notifications for Monday-side events (new leads, new to-dos)
<code>/api/cron/b2b-payment-check</code>	every 6 h	Confirm BECS settlement in MYOB → mark <code>payment_settled_at</code>
<code>/api/cron/crm-automations</code>	every 5 min	CRM automation flow engine
<code>/api/cron/task-automations</code>	every 5 min	Tasks automation flow engine
<code>/api/cron/crm-campaigns</code>	every 5 min	Campaign scheduler + Resend queue drain + call linkage
<code>/api/cron/ap-statement-watch</code>	every 10 min	Supplier statements → reconcile vs MYOB → digest email (report-only)
<code>/api/cron/ap-auto-entry</code>	every 15 min	VPS inbox → fact-check → auto-post clean invoices to MYOB (<code>AP_AUTO_ENTRY_ENABLED</code>)
<code>/api/cron/overnight-leads-snapshot</code> (+morning variant)	every 30 min / 5 min 06–08	Snapshot Monday lead groups for the sales recap
<code>/api/cron/letter-watch</code>	hourly	New finalised MYOB VPS invoices → thank-you letter + envelope print jobs (deposit-only invoices vetoed)
<code>/api/cron/agents</code>	every 15 min	Monitoring agents framework (<code>lib/agent-framework</code>)
<code>/api/cron/tune-jobs</code>	hourly :20	Scan inbox for Stripe tune receipts; Monday-morning reminders/escalations

Cron	Brisbane time	Purpose
/api/cron/b2b-dropship-confirm	every 15 min	Supplier confirmation emails → full drop-ship receiving flow
/api/cron/mgmt-dashboard-warm	05:30 daily	Pre-compute Management Dashboard MYOB bundles

⚠ Two handlers exist but are **not scheduled** (headers claim otherwise): `bank-payments-slack.ts` (7am payment digests) and `slack-cleanup.ts` (parts-bot TTL deletes). Manual-invoke only until added to `vercel.json`.

6.2 GitHub Actions (Mechanics Desk workers — 16 workflows)

MD has no API; these Playwright workers log in with `MECHANICDESK_{WORKSHOP_ID, USERNAME, PASSWORD}` secrets and talk back to the portal with `X-Service-Token: $JA_PORTAL_API_KEY`. All install Chromium at run time (`setup-node@v5` + `npx playwright install chromium` — the old "run in the MCR container" note in `05_INTEGRATIONS.md` is obsolete). **playwright in devDependencies must match the installed version (1.59.1)**. MD allows a single session per employee — concurrency groups prevent workers evicting each other.

Workflow	Schedule (Brisbane) / trigger	Job
mechanicdesk-pull	08/10/12/14/16:00	MD WIP report → forecast ingest
md-stock-sync	every 30 min, 06:00–19:30 Mon–Sat	Full MD stock catalogue → <code>md_stock_cache</code> (Slack parts bot)
md-workshop-map	03:30 nightly	Full invoices/quotes/customers pull → workshop map (FY2025 was a one-time backfill; nightly FROM = 2025-07-01)
mechanicdesk-prepick	06/11/15:00 weekdays	Diary-job parts demand → Pre Pick snapshot
sales-recap	Mon 07:00 full+email; ~7 intraday MD refreshes	Weekly Sales Recap (Monday boards + MD scrape → email Ryan)
md-customer-import	02:00 nightly	Monday quote-channel leads → MD customers (MD can't search phones)
tune-jobs-md	02:30 nightly	Submitted tune jobs → MD customer + vehicle + note
mechanicdesk-stocktake	repository_dispatch	Stocktake match / push / recheck / refresh / push-missing
md-purchase-order	repository_dispatch	Raise MD PO after a JAWS→VPS stock transfer
7 × probe-md-* / fix-md-customer-names	manual	Endpoint recon probes and one-off repairs

6.3 The print centre (comms room)

PORTAL-CENTRE is a dedicated always-on Lenovo ThinkCentre in the comms room whose only job is printing. It exists because the print agent used to run on Chris's MSI laptop, which meant **every letter, invoice and label silently stopped whenever the laptop was closed or off the network** — with no error anywhere, because the queue simply stopped draining. Built and verified 2026-08-20.

Host	Lenovo ThinkCentre, comms room, always on (no lid, no sleep)
Reachable	Tailscale <code>100.72.189.95</code> — RDP in for maintenance; it is Entra-joined, so sign in with the work account
Runs	<code>agents/label-print-agent/</code> on Node 22, started by Autologon + Startup shortcut, headless
Queue	<code>label_print_jobs</code> in Supabase, consumed over Realtime with a 30s poll as fallback
Job kinds	<code>label</code> (DYMO 4XL), <code>invoice</code> , <code>letter</code> , <code>envelope</code> (Apeos) — all four verified end to end
Runbook	<code>docs/print-centre-thinkcentre-setup.md</code> — build, headless operation, the comms-room move sequence, RDP, and a health-check script

Operating notes

- The agent **never restarts on deploy** — it is not part of the Vercel app. A new print kind must not require an agent change; add it as a queue row shape the agent already understands.
- A stuck queue drains on agent restart, so restarting the agent is the first fix. Failed jobs are re-queued by flipping `failed` → `pending`.
- ⚠ **Never trust the Apeos (Copy 1) printer-name suffix.** Windows appends it when a printer is re-added, and printing to the wrong name fails silently or prints to the wrong device — check the real name.
- Printer enumeration must never take the heartbeat down (fixed 2026-08-20) — if the agent looks dead but the box is up, check the heartbeat before rebuilding anything.
- DYMO troubles: the health-check script has label-printer diagnostics and a `-FixDymoPort` switch.

6.4 On-premise agents

Agent	Where	Notes
<code>label-print-agent</code> (<code>agents/label-print-agent/</code>)	PORTAL-CENTRE — see §6.3	Consumes <code>label_print_jobs</code> via Supabase Realtime (30s poll fallback). Kinds: <code>label</code> (DYMO 4XL), <code>invoice</code> , <code>letter</code> , <code>envelope</code> (Epson). Never restarts on deploy — new print kinds must not require agent changes. Runs via Startup shortcut → <code>run-agent-hidden.vbs</code> or NSSM. Node 22, Autologon, reachable on Tailscale <code>100.72.189.95</code> ; built 2026-08-20, runbook <code>docs/print-centre-thinkcentre-setup.md</code> . Previously ran on Chris's MSI laptop and failed silently whenever it was off-network — that is what the dedicated box fixes. Never trust the Apeos (Copy 1) printer-name suffix.
PBX workers	FreePBX box	See §5.10. systemd units; Node 16 for the camera bridges.
ESP32 scale nodes	Workshop bins / cash-count rig	Firmware <code>hardware/ja-scale-node/</code> ; raw HX711 counts to <code>/api/scales/ingest</code> ; all calibration server-side.

7. Modules & SOPs

Full route-by-route inventory: 111 page routes (423 API routes). Staff nav is the `/home` app launcher (role- and `visible_tabs`-filtered). Below, per module: what it is + how to operate it.

7.1 Dashboards (/dashboard , /overview , /home)

Section-based management dashboard (JAWS+VPS merged, entity filter pills) and a drag/resize custom dashboard builder with per-user saved layouts. **SOP**: nothing operational — data flows from MYOB reporting caches; if figures look stale, check the 02:00 `distributors/refresh-cache` cron and the health page.

7.2 Workshop tooling (Mechanics Desk stays the system of record)

Mechanics Desk is the workshop system. The diary, job cards, customers, vehicles, quotes and invoicing all happen in MD, and there is **no migration in progress** (decision 2026-08-20). The portal does not run the workshop.

What the portal *does* provide around MD, all of it in daily use:

Surface	What it does	Fed by
Letters (/workshop/letters)	Hourly <code>letter-watch</code> cron queues a thank-you letter + envelope for each newly finalised MYOB invoice and prints them on the Apeos. Deposit-only invoices are vetoed by a positive 1-1230 line. Reprint / compose / edit templates here.	MYOB (not MD)
Pre Pick (/workshop/prepick)	14-day parts demand vs stock on hand.	<code>mechanicdesk-prepick</code> Playwright worker
Parts & purchase orders (/workshop/orders , /workshop/purchase-orders , /workshop/suppliers)	Per-day parts worklist; PO draft → sent → received with a bill push to MYOB.	Portal + MYOB
Counting (/workshop/stocktake , Cash Count, Live Bins)	Portal-native barcode stocktake (session → count → Variance → Apply), till counting by weight, and HX711 load-cell bin inventory. Distinct from the MD stocktake at <code>/stocktake</code> (§7.12).	Portal / ESP32

Other `/workshop/*` routes exist in the codebase from the paused replacement build and are **not part of any operating procedure**. Leave them alone unless the migration is revived; `docs/workshop_md_parity.md` holds the parity and cutover detail if it ever is.

⚠ Because MD stays, the **9 scheduled MechanicDesk scrapers remain a live production dependency** (§6.2), not a temporary bridge. Treat them accordingly — an MD password change or UI change breaks real reporting.

7.3 B2B — distributor portal (/b2b/*)

Distributor experience: Shop → Cart (PO number required, ≤20 chars — MYOB limit) → Stripe Checkout (card+surcharge / PayTo / BECS) → Orders (status timeline + freight tracking) → Jobs (tune receipts to fill in) → Resources → Training → Team → Settings. Design language: the Alloy kit (`components/b2b/ui.tsx`) — distributor portal refreshed 2026-08-12, staff admin brought onto the same kit 2026-08-20.

What happens automatically on payment (`lib/b2b-order-pipeline.ts`): order → paid; cart cleared; MYOB invoice written cent-exact inc-GST; consignment-first pick list printed; for drop-ship items a supplier PO is created and emailed; Slack notification; freight bookable via MachShip (admin button or login-less email link).

The tax invoice is NOT raised at booking — see Freight & despatch below.

SOPs

- **Onboard a distributor:** Admin → B2B → Distributors → Add (live MYOB customer typeahead links the card) → open the distributor → Invite users (magic-link / welcome token email). Owners manage their own team after that.
- **Over-limit orders:** quantities above `over_limit_qty` route to a quote-or-dropship flow instead of straight checkout.
- **Bundles:** "includes" children ship inside the parent's box (affects freight cartonization).
- **Refunds:** admin order page → Refund modal → **Items mode** (pick lines + quantities; amount derived server-side from checkout-exact pricing; `refunded_qty` prevents double refunds) or amount modes (covers freight/surcharge). Mirrored to MYOB as a credit note with negative item lines (Xero path posts on B2B_SALES).
- **Drop-ship receiving:** supplier replies to the PO email → `b2b-dropship-confirm` cron reads it → auto-bills the PO, flips the sale order to invoice, receipts payment, relays ETA to the distributor, posts to `#jaws-orders`. Manual MYOB conversions are adopted rather than duplicated.
- **Freight & despatch (changed 2026-08-20 — read this):** booking and despatch are now two separate steps.
 - **Book Freight** only *prepares*: it creates the MachShip consignment (left **Unmanifested**) and prints the pick slip + consignment note/labels so the order can be picked and packed. Nothing reaches the carrier at this point and no tax invoice is raised.
 - **Ship Now** (`lib/b2b-ship-now.ts`) is what actually despatches: manifests the consignment, converts the MYOB Sale.Order → Sale.Invoice, receipts the payment against it, prints the A4 tax invoice, emails/pushes the distributor and stamps `shipped_at`.
 - Bulk despatch is deliberately **one manifest, not N** — MachShip's manifest call also books a carrier pickup window, so manifesting ten consignments individually would raise ten pickup requests. Select the run and ship it in one action.
 - This reverses the 2026-08-06 behaviour where booking manifested immediately. If nothing is reaching the carrier, the likely answer is simply that nobody has pressed Ship Now.
 - Rates: per-zone + flat-rate satchels (weight-gated; satchel rows may need seeding) + drop-ship per-zone rates; calibration panel at Admin → B2B → Dropship Calibration. If a consignment goes missing at MachShip it parks as `consignment_missing` — rebook from the order page.
- **Tune jobs:** Stripe receipt lands in the accounts inbox → hourly cron extracts it → distributor fills customer/vehicle at `/b2b/jobs` (or the login-less weekly reminder link) → nightly GH Action creates the MD customer/vehicle/note; Monday item + thank-you letter follow. One job per VIN. Staff-side management (aliases, retries, dismiss) at Admin → B2B → Tune Jobs.
- **Training:** Admin → B2B → Training → assign per distributor or per user (assignment-gated). Courses can be generated from an uploaded PDF (LLM pipeline). Edit quiz answers at `.../training/[slug]/answers`; preview renders the real player without recording attempts.
- **Testing safely:** Admin → B2B → Test Order exercises the full real pipeline against a chosen distributor.

7.4 B2B — staff admin (`/admin/b2b/*`)

Dashboard · Catalogue (inline price/visibility edits + drawer) · Stock Order (reorder forecasting, replaces the JAWS Excel) · JAWS Stocktake (count-sheet vs MYOB on-hand, **report-only**) · Stock Wall (saved on-hand tile views; also what suppliers see) · Stock Transfer (JAWS↔VPS paired invoice+bill + MD PO) · Distributors · Suppliers (logins) · Orders · Tune Jobs · Resources (sectioned doc library, signed uploads) · Training · Settings (Stripe status, freight carriers/zones/packaging, sender address, email templates).

7.5 Accounts Payable (/ap , /ap/[id] , /ap/statement)

Supplier emails → Graph inbox pull → Claude extraction → triage list.

Daily SOP (Amanda): /ap → Pull from Inbox if needed → review each invoice (/ap/[id] : PDF preview, line editor with account suggestions, MD job link, supplier presets) → Approve (pushes header + lines to the right MYOB entity) or Reject. Green-triage rows support bulk approve.

Automation: ap-auto-entry cron (VPS, gated by AP_AUTO_ENTRY_ENABLED) posts clean invoices automatically and Slacks a breakdown; supplier allowlists control consolidated and pay-on-proforma handling; duplicates get a 🔄 Slack and are filed to Read/Printed; **locked-period invoices are flagged, never auto-redated**; supplier matching is suffix-blind; link-only emails (no PDF attached) are invisible to the pipeline. ap-statement-watch cron reconciles statement PDFs against MYOB and emails a digest (report-only; Capricorn statements are report-only by policy). Manual statement recon UI: /ap/statement .

7.6 CRM (/crm/*)

Pipeline kanban, contacts (+timeline), campaigns (Resend), React Flow automations. Replaces Monday quote boards + ActiveCampaign + Zapier — **manual cutover steps are recorded in the project memory/notes; AC + Monday remain live in parallel until cutover**. Website leads arrive via /api/crm/intake (token-guarded). Three crons drive automations/campaigns/call-linkage every 5 min.

7.7 Calls (/calls)

CDR list with audio, transcripts, coaching analysis (per-call-type rubrics), Sentiment/Coaching/Words/Conversion tabs, live Listen/Whisper/Barge (needs monitor:calls), click-to-dial (use:phone , NEXT_PUBLIC_CLICK_TO_DIAL=1). Coaching attribution keys on the transcript-identified effective advisor. Weekly coaching report posts to Slack Monday 07:00. **Negative-call (concerns) automation is fully built but switched OFF** — CALL_CONCERNS_ENABLED=true resumes it.

SOP if calls stop appearing: check Connections page (freepbx_cdr_sync freshness) → SSH to the PBX via Tailscale → check ja-cdr-sync systemd timer. Transcripts stale → ja-transcribe . Live monitor "not configured" → ja-ami-monitor hasn't pushed a snapshot in >20s.

7.8 Reports (/reports/*)

Builder (7 PDF report types) · Sales Report (live Weekly Sales Recap; **"sales" = orders taken from Monday boards + MD, not turnover**; auto-emails Ryan Mon 07:00) · Management Dashboard (JAWS weekly Excel replica from live MYOB; config-driven charts; clickable KPI history; cache warmed 05:30) · Workshop Map (nightly MD pull; lib/workshop-map classification is authoritative; FY picker; five tabs — Jobs Map, Quotes Map, Conversion, By State, **Vehicle Trend**: one line per vehicle series, All FY = monthly buckets, pick a month = daily buckets, measures Jobs/Quotes/Job \$/Quoted \$. The trend counts every invoice and quote, so its totals run higher than the map tabs, which show one dot per customer per month) · Distributor Map (quotes near each distributor vs confirmed Monday bookings). Per-user report-tab allowlists control who sees what.

7.9 Tasks & Projects (/tasks , /projects , /todos)

Tasks: Monday-style board/list + React Flow automations (5-min cron). Projects: force-directed web of the six Monday "Hidden To Do" boards with comment posting back to Monday. To-Dos: manager scorecards over the same boards.

7.10 Messaging (/messages) & Notifications

Portal-native channels/DMs (Slack replacement for internal chat; WhatsApp/Messenger/IG phases not built). Notification bell + badges with 8 emitters; Web Push for staff and distributors (PWA installable).

7.11 Agents (/agents)

Monitoring-agent inbox (`lib/agent-framework` — not `lib/agents` , which is Pipeline A's mailbox map). Agents run every 15 min via cron; findings are triaged Dismiss / Mark done.

7.12 Stocktake — Mechanics Desk (/stocktake)

Upload count XLSX → Run Match (dispatches GH Action; includes coverage check) → review variance → Push to MD. Counted rows are sacred (row_number bug fixed 2026-07-30). MD is single-session — a stocktake run can 401 if someone logs into MD as the same employee mid-run (known, deferred).

7.13 Stripe→MYOB (/stripe-myob)

Lists Stripe invoices per account label and pushes them to MYOB as Professional Invoice + Customer Payment pairs; payout reconciliation endpoints support the JAWS accounts.

7.14 Sales/analysis surfaces

`/distributors` (group-aware revenue reporting; groups managed at `/admin/groups`), `/sales` , `/stock` (⚠ no SSR gate), `/forecasting` (MD job report forecast; `/jobs` redirects here), `/vehicle-sales` (platform classification cache from VPS invoices), `/job-reports` (MD job report ingest for PO→job matching).

7.15 Settings (/settings)

Tile launcher: General · Connections (Integrations / Health / MYOB) · Distributor Report config · VIN Codes · Users · Audit log · Profile · Claude connector (MCP tokens) · Service tokens · **Library**.

7.16 Library (/admin/library)

This document and the SOP, served inside the portal — readable on screen with a contents rail, and downloadable as PDF. Registry: `lib/library-docs.ts` (one row per document). Gated on `admin:settings` .

- `docs/*.md` is the **source of truth**; the reader renders it live server-side (`marked`), so editing the markdown updates the on-screen copy at the next deploy.
- `docs/*.pdf` is a **generated artifact** — regenerate with `scripts/render-doc-pdf.js` (marked + Playwright; needs `npm install playwright chromium` once) and commit it. It goes stale silently otherwise.
- ⚠ **The PDFs are deliberately not in `public/`** . This document names where every credential lives, the Supabase project id, the Tailscale addresses and the open security gaps; anything under `public/` is served to the internet to anyone holding the URL, with no sign-in. Downloads go through `/api/admin/library/[slug]` , which is auth-gated.
- ⚠ `next.config.js` → `outputFileTracingIncludes` **must list `docs/**` for the three Library routes**. The paths are built at runtime so Next's tracer can't see them; without it the files are pruned from the serverless bundle and 404 in production while working perfectly in dev.

SOP: see `CLAUDE.md` §1 — every change to the portal updates these documents as part of the same work.

8. Monitoring & troubleshooting

1. **First stop:** `/admin/connections` (or Settings → Connections → Health). 21 checks across accounting / workshop / comms / crm / phone / infra, written every 5 min by the health cron. Manual force: `curl -H "Authorization: Bearer $CRON_SECRET" https://justautos.app/api/cron/health-check?force=1`.
 2. **Freshness-based checks** (PBX CDR sync, transcribe, Deepgram, MD pulls) go red when data stops arriving — the fix is on the source host/worker, not the portal.
 3. **Graph mailbox rows red** → renewal cron failed or the subscription died: re-run `setup-graph-subscriptions` (idempotent).
 4. **MYOB rows red** → token lapsed: re-run the connect flow (Settings → MYOB Connection). Remember refresh tokens die if unused for weeks.
 5. **GH Actions rows red** → check the Actions tab on the repo; MD workers upload failure artifacts and Slack on failure.
 6. **A B2B order paid but no MYOB invoice** → `b2b_orders.myob_write_error`; fix cause, retry from the admin order page. Stripe webhook itself is idempotent.
 7. **Prints not coming out** → the workshop PC agent: check the tray/NSSM service; it drains pending jobs on startup, so restarting it usually clears the backlog.
-

9. Known risks, debt & outstanding items (as of 2026-08-20)

Security

- Supabase `service_role` key was exposed in an April 2026 session and **has never been rotated**. Rotation must update: Vercel env, the FreePBX host workers, the workshop print agent `.env`, and GitHub Actions secrets — all hold it.
- Repo working tree contains `agents/label-print-agent/.env` and `hardware/ja-scale-node/secrets.h` — verify both are gitignored and never committed with live values.
- `NEXT_PUBLIC_GOOGLE_PLACES_API_KEY` is client-exposed — must stay referrer-restricted.
- Three pages lack SSR auth gates (`/stock`, `/imports`, `/admin/connections`) — API-level gating covers data but the pages render.

Consistency / drift

- `lib/auth.ts` role type is missing `workshop` (use `lib/permissions.ts`).
- Migrations `148` and `153` are duplicated numbers; latest applied is `196_md_vehicle_trend`, so next is `197`.
- `bank-payments-slack` and `slack-cleanup` cron handlers exist but aren't scheduled in `vercel.json`.
- Seven overlapping base-URL env vars.
- Older docs (`03`, `05`, `07`) predate the CData decommission, B2B rollout and Xero foundation, and still describe the workshop migration as active — trust this document and the code.

Operational

- MYOB and Xero OAuth tokens need periodic exercise; Xero refresh tokens are single-use (rotation is handled, but never hand-edit the row).

- The FreePBX box is CentOS 7 — camera bridges pinned to Node 16; the whole host is a single point of failure for calls, coaching, and camera alerts. `sip-loss-monitor.sh` may still be running there from the phone-dropout investigation (upstream/FreedomBroadband was the cause) — clean up when resolved.
- **MechanicDesk is staying** (decision 2026-08-20) — the replacement build is paused, so the 9 scheduled MD scrapers are a permanent production dependency rather than a temporary bridge. The portal's workshop module is built but unused; `docs/workshop_md_parity.md` keeps the cutover checklist if it is ever revived.
- Parked/off: negative-call automation (`CALL_CONCERNS_ENABLED`), portal-side calls analysis cron, Places key on New Booking quick-add (key unset), first real JAWS→VPS stock transfer pending, Live Bins hardware untested in production, live call monitoring WSS keep-alive retest pending.
- MYOB→Xero migration: foundation only — adapter waves per module still to come; both entities eventually move.

10. Key file index

Area	Files
Auth/permissions	<code>lib/permissions.ts</code> , <code>lib/authServer.ts</code> , <code>lib/auth.ts</code> , <code>lib/b2bAuthServer.ts</code> , <code>lib/b2bSupplierAuth.ts</code> , <code>lib/service-auth.ts</code>
Credential resolver	<code>lib/integration-config.ts</code> , <code>pages/api/admin/integrations.ts</code> , <code>components/settings/IntegrationsTab.tsx</code>
Accounting	<code>lib/myob.ts</code> , <code>lib/myob-reporting.ts</code> , <code>lib/xero.ts</code> , <code>lib/accounting-provider.ts</code> , <code>lib/accounting/*</code>
B2B pipeline	<code>lib/b2b-order-pipeline.ts</code> , <code>lib/b2b-payment.ts</code> , <code>lib/b2b-myob-invoice.ts</code> , <code>lib/b2b-machship.ts</code> , <code>lib/b2b-freight-*.ts</code> , <code>lib/b2b-dropship-*.ts</code> , <code>lib/b2b-tune-jobs.ts</code>
AP	<code>lib/ap-extraction.ts</code> , <code>lib/ap-auto-entry.ts</code> , <code>lib/ap-statement-watch.ts</code> , <code>lib/ap-myob-bill.ts</code>
Workshop tooling	<code>pages/workshop/letters</code> , <code>.../prepick</code> , <code>.../purchase-orders</code> , <code>.../stocktake</code> , <code>lib/workshop-*.ts</code> . Paused replacement build: rest of <code>pages/workshop/*</code> + <code>docs/workshop_md_parity.md</code>
Calls	<code>lib/live-calls.ts</code> , <code>lib/calls-analysis.ts</code> , <code>lib/softphone.ts</code> , <code>docs/pbx_click_to_dial_worker.md</code>
Mail	<code>lib/microsoft-graph.ts</code> , <code>lib/email.ts</code> , <code>lib/agents.ts</code> (mailbox→owner map)
Health	<code>pages/api/cron/health-check.ts</code> , <code>pages/admin/connections.tsx</code>
Workers	<code>.github/workflows/*</code> , <code>scripts/*</code> , <code>agents/label-print-agent/</code> , <code>agents/ja-freightbay/</code> , <code>hardware/ja-scale-node/</code>
Config	<code>vercel.json</code> (crons + maxDurations), <code>migrations/</code>